ALAN MARTINS

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Github Linkedin

Summary

Senior Full Stack Developer and AI Specialist with deep expertise in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Generative AI. Skilled in building and deploying intelligent, scalable systems using LangChain, Python, OpenAI, and cloud architectures (AWS/GCP). Combines hands-on engineering, product thinking, and a data-driven mindset to deliver high-impact AI solutions from concept to production.

Experience

11/2024 - Present

Backend Development Specialized in LLMs

Radix - *Rio de Janeiro, Rio de Janeiro*

- Implementation of RAG (Retrieval-Augmented Generation) pipelines with vector databases such as MongoDB Atlas Vector Search and Pinecone.
- Fine-tuning and transfer learning of LLMs for specialized NLP tasks.
- Development of intelligent agents using LangChain and OpenAI Assistants API.
- Natural Language Processing (NLP) for classification, summarization, and automated content generation.
- Experiments with Convolutional (CNN) and Recurrent Neural Networks (RNN) for computer vision and time series tasks.
- Developed scalable AI pipelines using MERN stack with focus on high-volume performance.
- Implemented advanced RAG systems using LangChain and vector databases (MongoDB Vector, Pinecone).

01/2022 - 10/2024

Backend Development Specialized in LLMs

Grupo Urbam - *Rio de Janeiro, RJ*

- Specialized in LLMs and LangChain to build AI-driven solutions with real-world impact.
- Developed scalable backend systems using Python and Node.js, integrated with cloud services (AWS: EC2, S3, Lambda, CodePipeline; GCP).
- Designed and implemented RAG (Retrieval-Augmented Generation) pipelines to improve the accuracy and contextual relevance of AI responses.
- Created a computer vision solution integrated into a mobile app for sanitation workers to document cleanliness via photo capture.
- Applied few-shot learning with LLMs to evaluate image context and determine environmental cleanliness based on visual and textual data.
- Resulted in a 12% reduction in customer complaints by automating and improving the accuracy of site condition assessments.

02/2016 - Present

Instructor

Step Computer Academy Brasil - *Rio de Janeiro, RJ*

- Extensive experience in the mobile application development lifecycle, from prototyping and UI/UX design to deployment on app stores (App Store and Google Play).
- Delivered practical workshops and classes on Next.js, React Native, Expo, and MongoDB, combining foundational theory with real-world application.
- Developed hands-on experiments using agile methodologies and frameworks such as Scrum.
- Skilled in engaging teams and enhancing technical learning through strategic coaching and collaborative development practices.
- Experienced with modern DevOps practices, including Docker Compose, Kubernetes, CI/CD pipelines, and cloud-based infrastructure automation.
- Solid foundation in Software Engineering principles, focusing on maintainability, scalability, and clean architecture.

02/2010 - 01/2017

Desenvolvedor Sênior Full Stack
Azimute startup - *Rio de Janeiro, RJ*

- Developed mobile applications with React Native and Expo, ensuring high performance and seamless user experience.
- Integrated with banking APIs to implement secure and robust communication and financial transaction features.
- Published and managed apps on the App Store and Google Play.
- Modeled and architected applications with a focus on scalability and efficient maintenance.
- Created APIs using Node.js and Python, optimized for high performance and security.
- Utilized Google Cloud Platform (GCP) for scalable hosting and infrastructure services.
- Implemented backend functionalities and authentication with Firebase.
- Collaborated with cross-functional teams, following agile methodologies for continuous value delivery.

Skills

- Python
- OpenAI API / Assistants API
- TypeScript
- LangChain
- Next.Js
- Prompt Engineering
- Embeddings
- AWS
- Google Cloud
- Docker
- RAG
- Few-shot learning
- CI/CD
- Github
- Nest.Js
- LLM
- Doker
- TensorFlow
- React Native
- SOLID Principles
- DDD (Domain-Driven Design)
- Agile / Scrum

Education

01/2024 - 10/2025

Postgraduate Program in Artificial Intelligence - UFV - Viçosa, MG
Status - Currently Pursuing
A course focused on machine learning, neural networks, computer vision, and natural language processing. Developed skills to solve real-world problems and design intelligent systems, with an emphasis on research and practical applications.

01/2018 - 01/2022

Technologist in Systems Analysis and Development - FAETERJ - *Rio de Janeiro, RJ*
Status - *Graduated*
Education focused on system development and maintenance, with expertise in programming, databases, and software engineering tailored to IT market demands.

01/2023 - 05/2025

UFRJ **- Master's in Data Science with a Focus on Education**
Status - Currently Pursuing
An interdisciplinary program that applies data science to address educational challenges, emphasizing personalized learning, performance analysis, and technological innovation in

01/2012 - 02/2013

Mechatronics Technician ***- Rezende Ramel*** - *Rio de Janeiro, RJ*
Status - *Graduated*
Mechatronics Technician
A technical course that combines mechanics, electronics, and computer science to develop and maintain automated systems and robotics. It prepares professionals for industrial automation, system optimization, and electronic device control.

Languages

Português: Native language

Inglês: B1

Pré-intermediário

Projects & Certifications

- AWS Certified Cloud Practitioner: Certification in cloud computing fundamentals, covering AWS core services, security, and best practices.
- EF SET English Certificate: Verified English proficiency with an internationally recognized certification.
- Brain IA Legal Assistant, App jurídico automatizado usando IA, MongoDBVector, LangChain e RAG.
- HackPUC 2017, Finalista em maratona de programação.
- Conjuntos APP, Apresentado no 8º CONEDU.